

Tyler King's Background

A. Personal Story

African-American Protagonist: Tyler represents a young Black innovator, emphasizing resilience, ingenuity, and familial dedication.

Family Dynamics:

Grandparents: Tyler is caring for his aging grandparents, creating stakes and urgency for his success.

Father: An active figure, but physically distant, based in Kenya. His father's work at the Raspberry Pi facility inspires Tyler's use of the technology.

Cultural Identity:

The game explores Tyler's dual identity as an African-American and the influence of his father's Kenyan heritage. This could include subtle cultural references in his interactions, storytelling, or even the design of his creations.

B. Motivation

Tyler's ingenuity stems from necessity: his family's financial situation and the need to support his grandparents during the pandemic.

His father's encouragement and the Raspberry Pi gift serve as a catalyst for Tyler's journey, symbolizing both connection and empowerment.

2. Interactive Intelligence: AI Companion

A. The AI Assistant

Inspired by Navi (Zelda): The AI assistant, sent by Tyler's father, serves as an interactive intelligence guiding him through challenges.

Role in the Game:

Provides hints and real-time suggestions.

Offers a status window mechanic similar to solo leveling or lone necromancer:

Displays Tyler's skill progression.

Tracks system health for FarmBot, solar panels, composting, and other devices.

Monitors resources (e.g., electricity generated, water recycled, food grown).

Personality:

The AI is not just a tool but a character, blending technical expertise with emotional support.

It can reflect Tyler's father's teachings and philosophy, providing a sense of his presence even across continents.

B. Integration with Raspberry Pi

The Brain of the System: The Raspberry Pi powers not only the FarmBot but also the custom AI assistant.

Endgame Connection:

Tyler's backyard project inspires his father to pitch a similar system for Raspberry Pi's operations in Kenya.

The game closes with a message about global circular economy initiatives and Tyler's contribution to a larger movement.

3. Gameplay Mechanics: Status Window and Decision Tree

A. Status Window

Skill Progression:

Tracks Tyler's growth in engineering, robotics, and sustainability.

Skills are visualized as a tree or dashboard, showing current levels and unlockable abilities.

System Health:

Monitors the efficiency and functionality of his circular systems.

Alerts for repairs, upgrades, or resource shortages.

B. Decision Tree

Every choice Tyler makes has consequences, shaping his progress and the systems he builds:

Example:

Invest time upgrading the composting system for better soil, or focus on repairing the solar panels to ensure power.

Choices affect the sustainability of the backyard and Tyler's ability to meet his family's needs.

4. Narrative Depth

A. Themes of Resilience and Circularity

The story centers on making the most out of limited resources, mirroring real-world issues like waste management and environmental sustainability.

Tyler's ingenuity in transforming "waste" into valuable systems becomes a metaphor for Black excellence and innovation under pressure.

B. Father-Son Connection

Tyler's father:

Sends Tyler the Raspberry Pi and AI operating system as a symbol of their shared passion for technology.

Acts as a remote mentor, offering advice through the AI system he designed.

The game can include flashback sequences or voice messages from Tyler's father, adding emotional weight.

C. Social Commentary

The nearby dump reflects real-world issues of consumption and waste culture.

Tyler's success highlights the potential of circular economies to empower individuals and communities, especially those marginalized by traditional systems.

5. Broader Implications: Expanding Tyler's World

A. The Endgame

Tyler's backyard system becomes a prototype for larger-scale implementations, bridging local sustainability with global impact.

The narrative can lead to partnerships with schools or organizations in Kenya, reflecting his father's work and Tyler's growth.

B. Expansions or Sequels

Introduce multiplayer modes or additional characters inspired by other regions or challenges, expanding the world of FarmBot Circularity.

Include collaborative missions where players build circular systems for communities.

6. Realizing the Project: Development Plan

A. Game Framework

Start with a Scratch prototype:

Focus on the AI assistant, status window, and crafting mechanics.

Use HTML5 and JavaScript for the final version, incorporating animations and interactive elements.

Design the story and characters with input from diverse cultural consultants to ensure authentic representation.

B. Funding Opportunities

Apply for grants like 4Culture First or STEM education funding.

Collaborate with Raspberry Pi, FarmBot, or sustainability organizations for support.

C. Scalability

Develop educational modules based on the game's mechanics.

License the game to schools and libraries as a tool for STEM education.

7. Conclusion: A Legacy of Innovation

This game combines circular economy principles, Black representation, and STEM education in a way that is both meaningful and marketable. It's a showcase of your ability to merge storytelling with technical expertise, while addressing real-world challenges through an interactive medium.

Would you like help drafting the storyline, creating the Scratch prototype, or planning outreach to partners like Raspberry Pi or 4Culture?